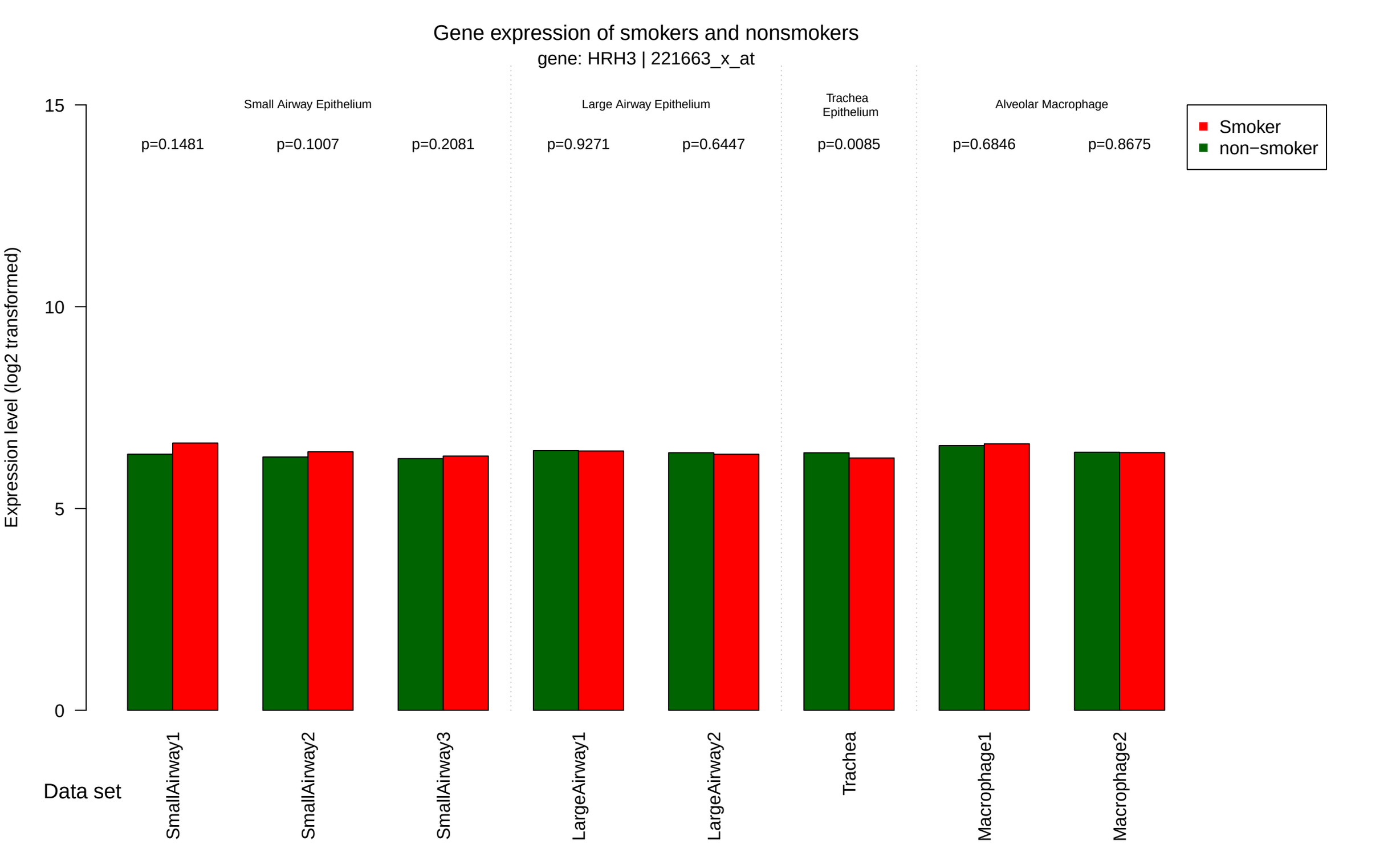
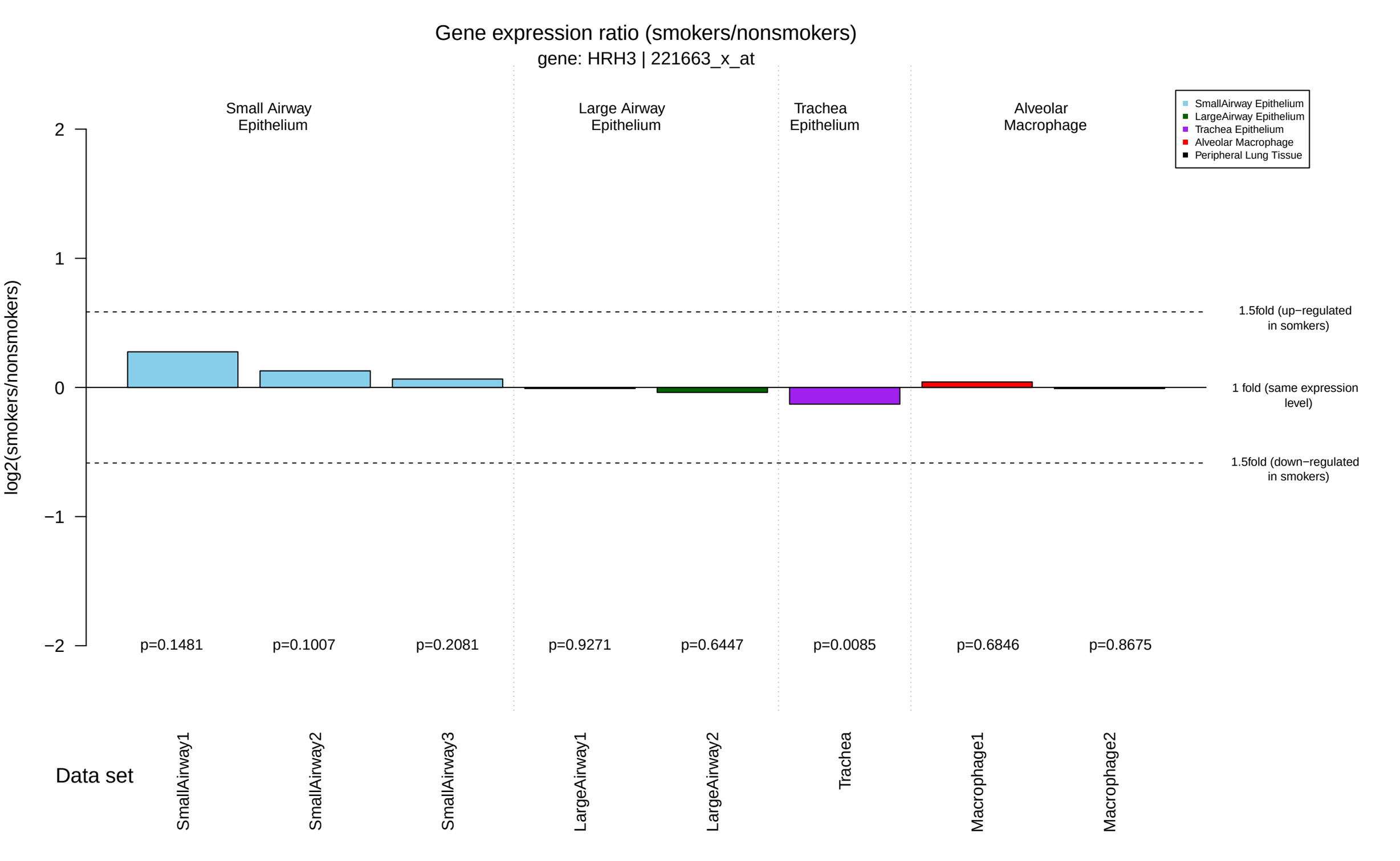
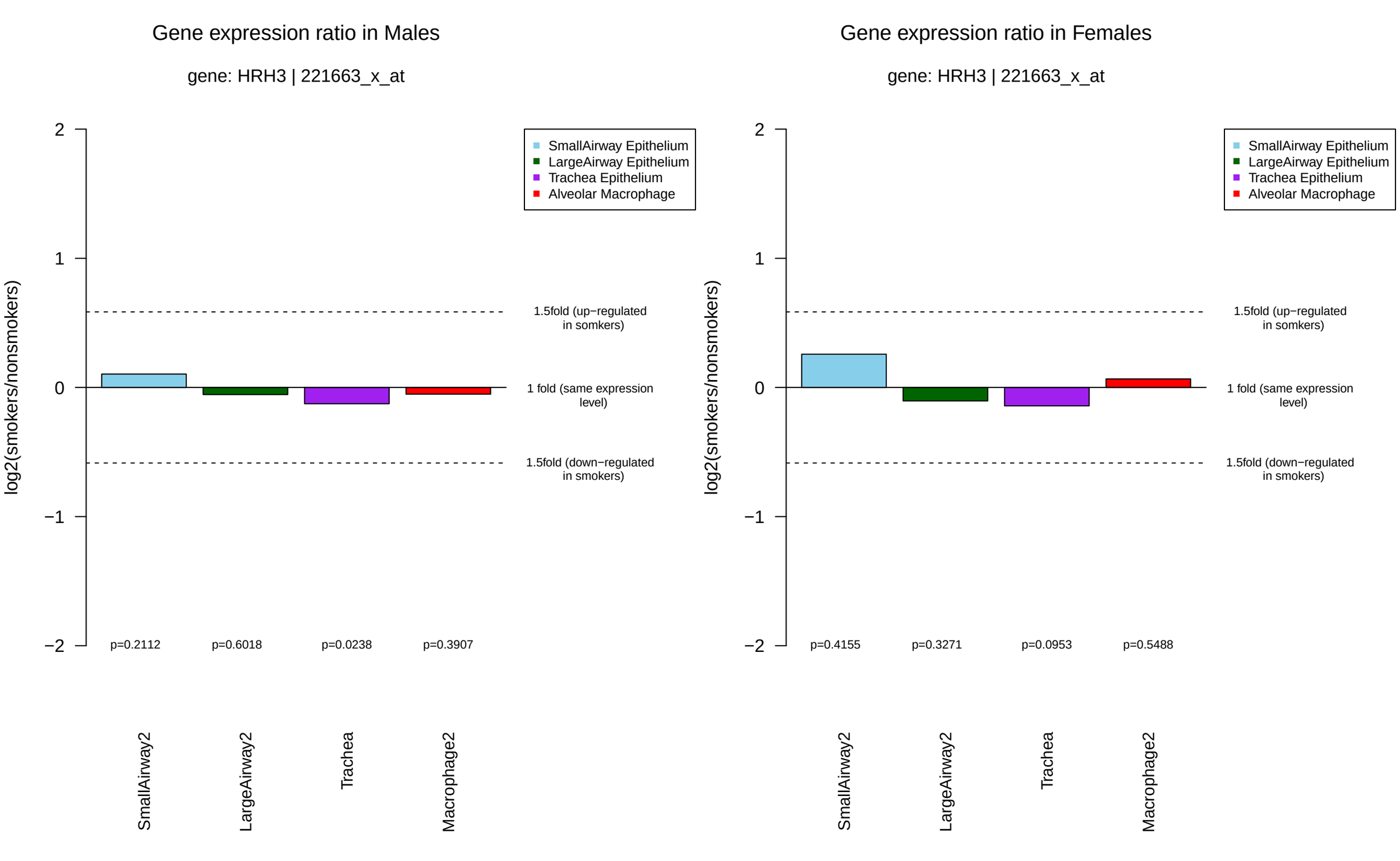
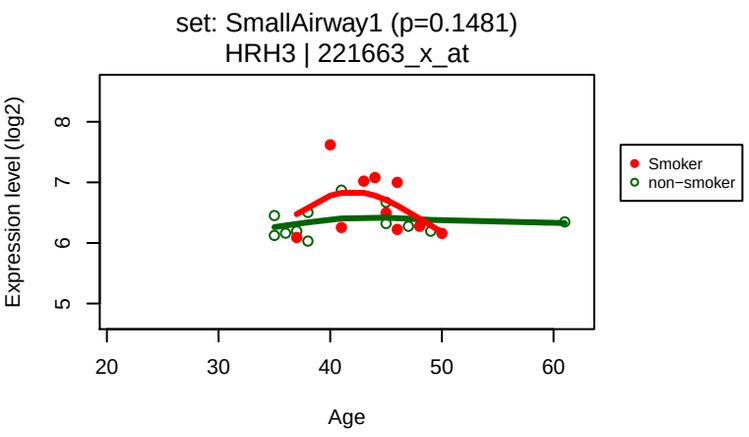
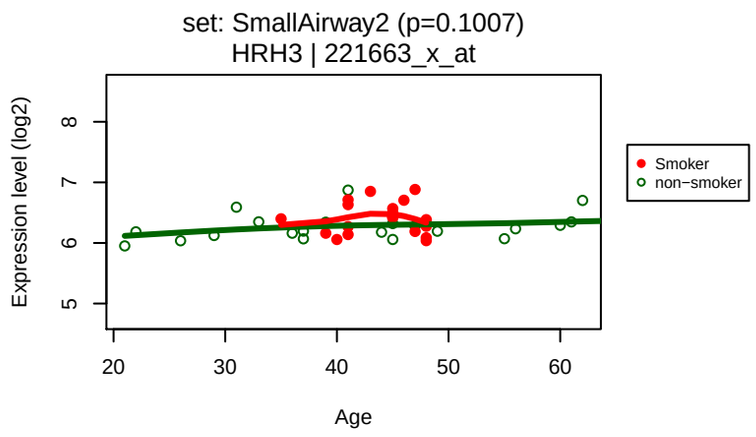
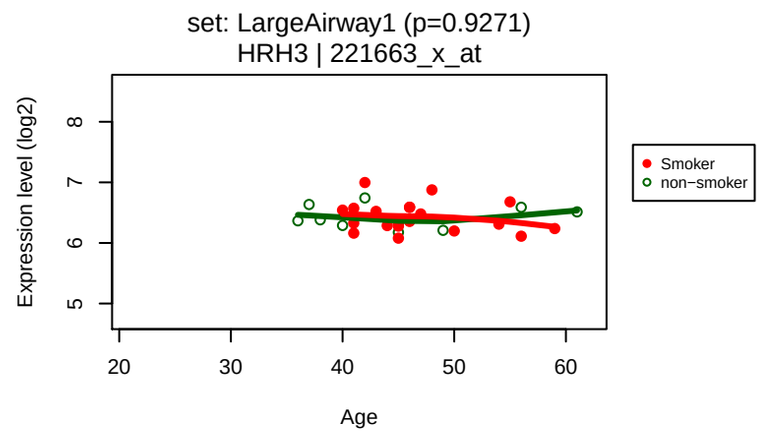

Gene expression of smokers and nonsmokers
gene: HRH3 | 221663_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1481
p=0.1007
p=0.2081
p=0.9271
p=0.6447
p=0.0085
p=0.6846
p=0.8675
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

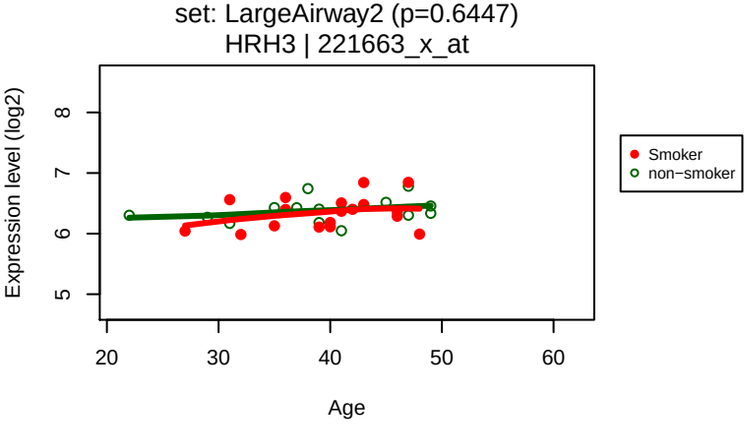

Gene expression ratio (smokers/nonsmokers)
gene: HRH3 | 221663_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1481
p=0.1007
p=0.2081
p=0.9271
p=0.6447
p=0.0085
p=0.6846
p=0.8675
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

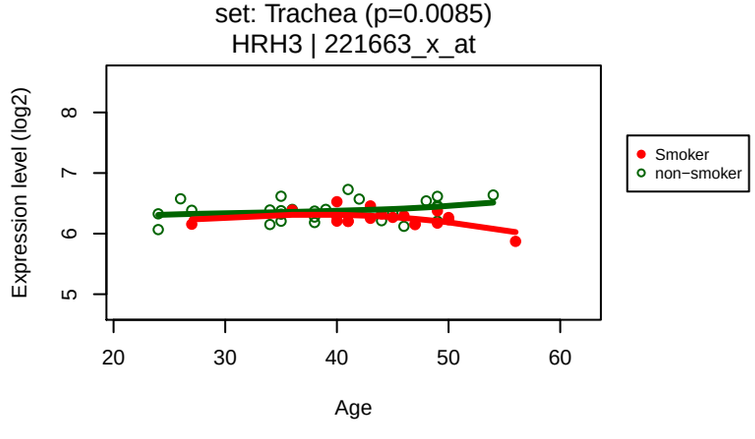

Gene expression ratio in Males
gene: HRH3 | 221663_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2112
p=0.6018
p=0.0238
p=0.3907
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HRH3 | 221663_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4155
p=0.3271
p=0.0953
p=0.5488
SmallAirway2
LargeAirway2
Trachea
Macrophage2

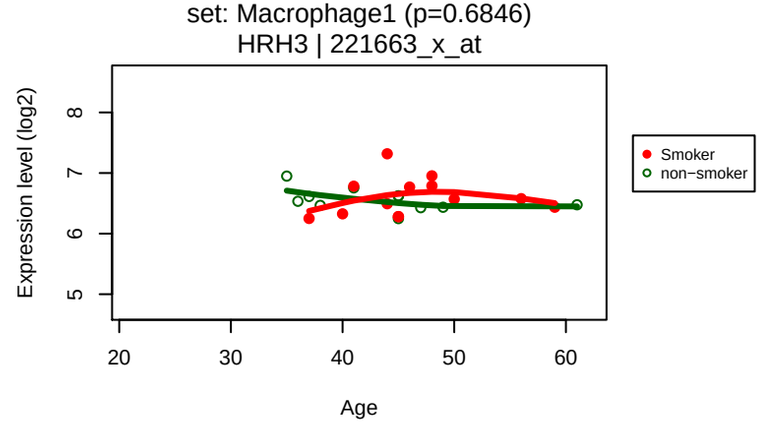

set: SmallAirway1 (p=0.1481)
HRH3 | 221663_x_at
Expression level (log2)
Age
Smoker
non−smoker

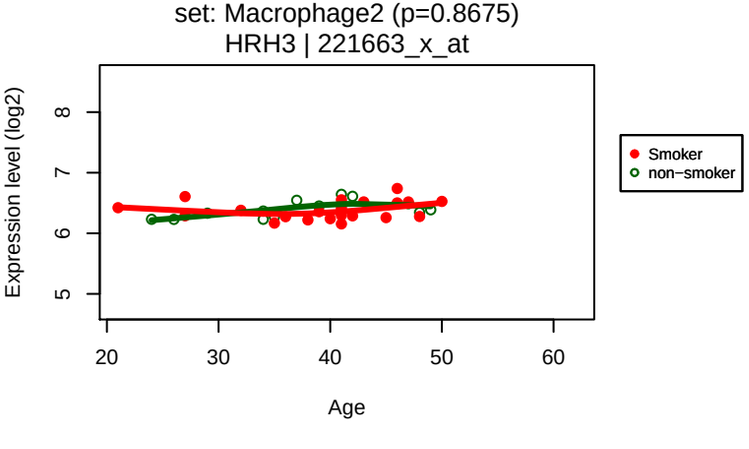

set: SmallAirway2 (p=0.1007)
HRH3 | 221663_x_at
Expression level (log2)
Age
Smoker
non−smoker

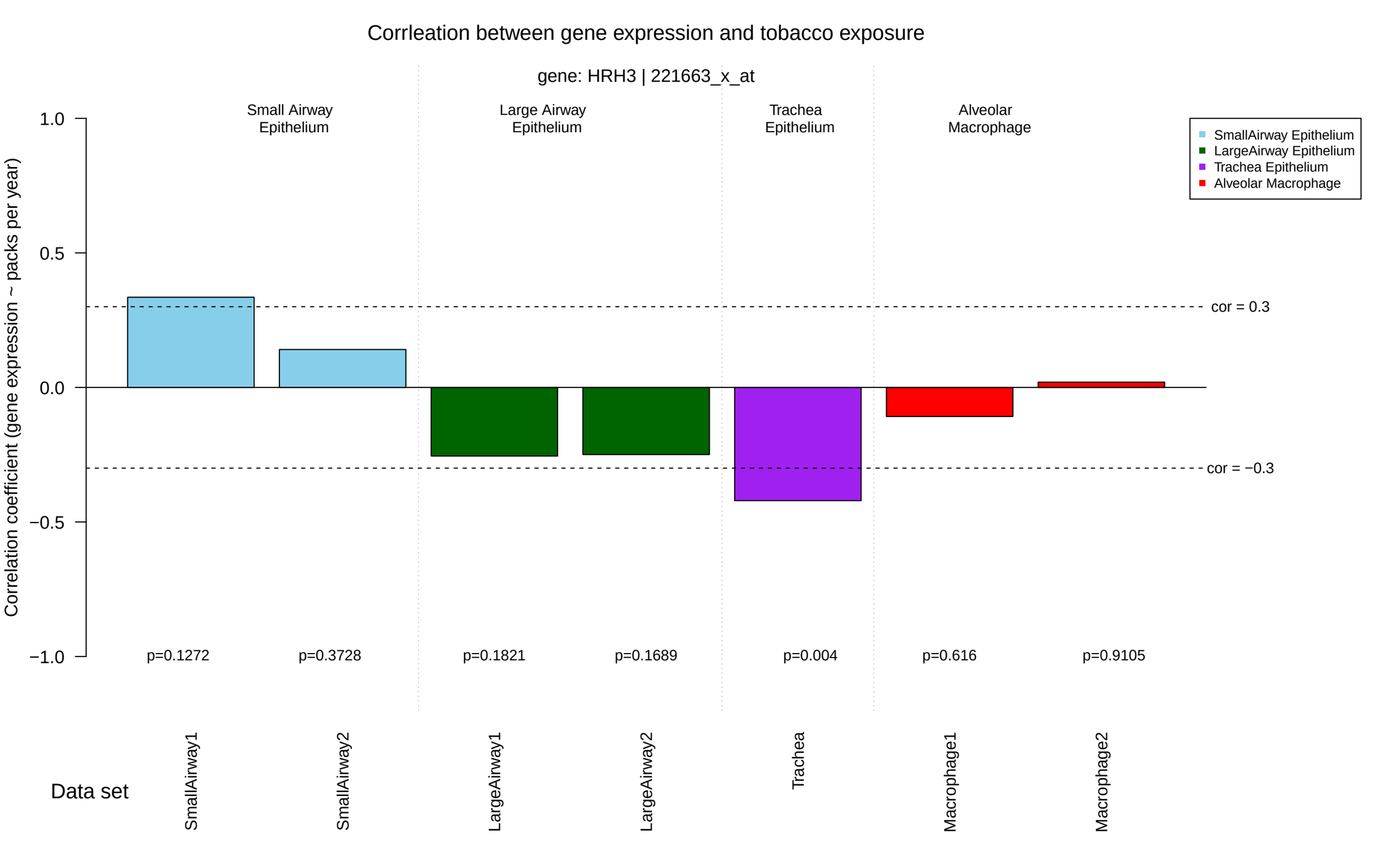

set: LargeAirway1 (p=0.9271)
HRH3 | 221663_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6447)
HRH3 | 221663_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0085)
HRH3 | 221663_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6846)
HRH3 | 221663_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8675)
HRH3 | 221663_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HRH3 | 221663_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1272
p=0.3728
p=0.1821
p=0.1689
p=0.004
p=0.616
p=0.9105
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2